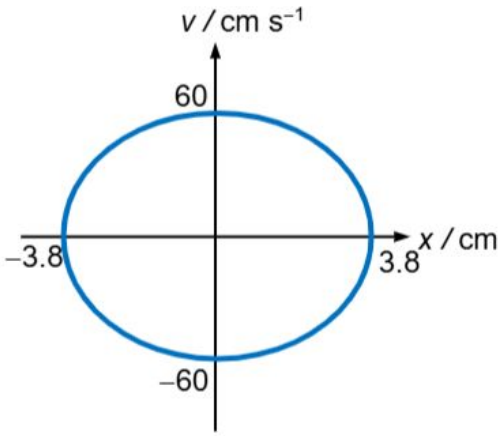
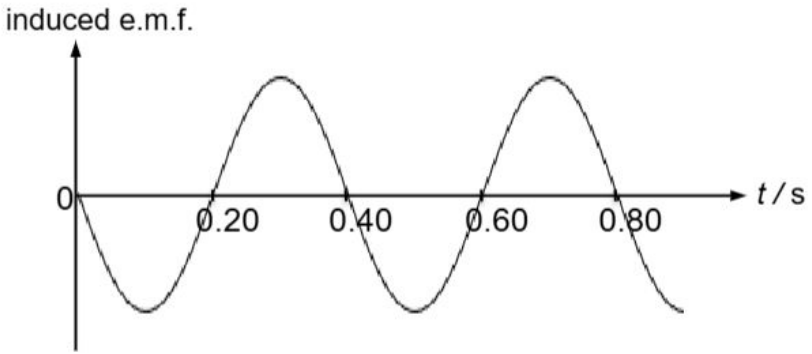
6	
(a)	The resultant force / acceleration is always directed toward the equilibrium position/fixed point
	The resultant force / acceleration is directly proportional to the displacement / distance from the equilibrium positon/fixed point.
(b)(i)1.	$\omega = \dfrac{2\pi}{T}$ $= \dfrac{2\pi}{0.40}$
	$= 15.7 \text{ rad s}^{-1}$
(b)(i)2.	When displacement is zero, acceleration = 0
	When displacement is maximum, $a = \omega^2 x_0$ $= 15.708^2 \times 0.038$
	$= 9.3761$ $= 9.38 \text{ m s}^{-1}$
(b)(ii)	
	Shape of graph
	Labelling of *x*-axis and *v*-axis with values
(b)(iii)	$v = \omega\sqrt{x_0^2 - x^2}$ $\frac{1}{2}\omega x_0 = \omega\sqrt{x_0^2 - x^2}$
	$0.25x_0^2 = x_0^2 - x^2$ $x = \left(\sqrt{0.75}\right)(3.8)$
	$= 3.29 \text{ cm}$
	Alternatively, $x = -x_0 \cos \omega t$ $v = v_0 \sin \omega t$ $\dfrac{1}{2} v_0 = v_0 \sin \omega t$
	$t = 0.0333$ s
	$x = 3.8\cos(15.7 \times 0.0333)$ $= 3.29 \text{ cm}$

(c)(i)	
	Shape of graph: sine/-sine
	Correct period
(c)(ii)	At the top or bottom of the oscillation, the magnet is momentarily at rest and hence, there is no change in magnetic flux linkage and there is no e.m.f. produced.
	At the equilibrium position, the velocity of the magnet is maximum and hence rate of change of flux linkage (or rate of cutting of flux) is a maximum. According to Faraday's Law, the induced e.m.f. is maximum.
	According to Lenz's Law, when the magnetic flux linkage increases, the induced e.m.f. will be such that the induced current if it flows will be such as to oppose the increase in magnetic flux linkage. Hence the induced e.m.f. have opposite signs when the magnet is moving down and when it is moving up.
(c)(iii)	(When the resistor is connected across XY, the circuit is closed.) There will now be an induced current flowing in the coil,
	producing a magnetic force that opposes the oscillation / energy is dissipated as thermal energy in the coil.
	Hence the amplitude of the oscillations decreases continuously over time.
7	
(a)(i)	$z = 92 - 42 = 50$